Status	Finished
Started	Thursday, 29 October 2026, 6:07 PM
Completed	Thursday, 29 October 2026, 6:14 PM
Duration	7 mins 58 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By breaking the input into smaller parts and merging them
- b. By sorting data in a single pass
- c. By comparing elements sequentially
- d. By using the bubble-up method

The correct answer is: By breaking the input into smaller parts and merging them

Question **2**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Sorting
- b. Merging
- c. Rearranging
- d. Searching

The correct answer is: Sorting

Question **3**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Inserting data into a list
b. Deleting data from a list
c. Arranging data in a particular format
d. Searching for data in a list

The correct answer is: Arranging data in a particular format

Question **4**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It avoids the need for recursion
b. It simplifies the sorting process by using only one pass
c. It allows for efficient parallel processing and sorting of data
d. It only works on small datasets

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **5**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Makes data harder to manage
b. Provides a basis for other algorithms to work efficiently
c. Makes code execution slower
d. Decreases the efficiency of algorithms

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

- a. Merge Sort
- b. Bubble Sort
- c. Quick Sort
- d. Insertion Sort

The correct answer is: Merge Sort

Question **7**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Sorting the entire list in one pass
- b. Dividing the list into halves
- c. Bubbling up the smallest element
- d. Bubbling up the largest element to its correct position

The correct answer is: Bubbling up the largest element to its correct position

Question **8**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting makes it harder to search for items
- b. Sorting helps in finding duplicates quickly
- c. Sorting is rarely used in programming
- d. Sorting decreases the efficiency of selection operations

The correct answer is: Sorting helps in finding duplicates quickly

Question **9**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Dividing the input into parts, solving each part, and combining the solutions
- b. Rearranging data without sorting
- c. Sorting data sequentially
- d. Sorting data in a single pass

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **10**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It makes it easier to select items based on their relationship to the rest of the items
- b. It makes the data unsorted
- c. It slows down the process
- d. It complicates the selection of items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **11**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort is less efficient for large lists
- b. Bubble Sort cannot handle duplicate elements
- c. Bubble Sort is difficult to implement
- d. Bubble Sort does not guarantee sorted order

The correct answer is: Bubble Sort is less efficient for large lists

Question **12**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

- a. A sorting algorithm is stable if it preserves the order of duplicate keys
- b. A sorting algorithm is stable if it preserves the order of non-duplicate keys
- c. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
- d. A sorting algorithm is stable if it preserves the order of all keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **13**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It builds a sorted array one element at a time
- b. It compares adjacent elements and swaps them if necessary
- c. It divides the list into two halves, sorts each half, and then merges them
- d. It repeatedly finds the minimum element and moves it to the sorted part of the list

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **14**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Selection
- b. Insertion
- c. Extraction
- d. Distribution

The correct answer is:
Insertion

Question **15**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. sorted()
- b. sort()
- c. arrange()
- d. order()

The correct answer is: sort()